System Manual: Apex P200 Industrial Articulated Robot

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1. System Overview & Core Specifications

The Apex P200 is a high-speed, 6-axis industrial articulated robot manipulator engineered for precision assembly, pick-and-place, and heavy-duty robotic welding. The mechanical arm incorporates absolute optical encoders on all six joints with battery-backed position retention during main supply interruptions.

Parameter	Design Specification	Tolerance / Unit
Maximum Payload Capacity	25.0 kg	± 0.2 kg nominal
Maximum Reach Envelope	1,850 mm	Spherical radial reach
Pose Repeatability (ISO 9283)	± 0.03 mm	Tested at full extension
Robot Weight (Mechanical)	235 kg	Excluding base pedestal
Operating Voltage	400 V AC ±10%	3-Phase, 50/60 Hz
Total Power Consumption	4.2 kVA	At continuous maximum duty
Ambient Operating Temp	0°C to +45°C	Relative humidity 5% to 85%
IP Ingress Protection	IP67 Wrist / IP54 Body	Standard factory configuration

2. Safety Systems & Emergency Stop Architecture

All P200 installations must comply with ISO 10218-1 and ANSI/RIA R15.06 industrial safety standards. The emergency stop safety loop is designed with a redundant dual-channel Category 4 / Performance Level e (PL e) architecture.

2.1 Emergency Stop Specifications & Trip Conditions

The primary E-Stop circuit utilizes safety contactors K1 and K2 wired to redundant terminal blocks TB1-12 and TB1-14. The total emergency stop response time from button actuation to zero mechanical movement must NOT exceed 25 milliseconds. Mechanical brakes engage automatically on all 6 axes within 15 milliseconds of power bus disconnection.

2.2 LiPo Backup Battery Replacement Protocol

The robot base pedestal houses a 3.6V lithium thionyl chloride (Li-SOCl2) backup battery pack (Part No. BAT-36-P200). This battery maintains absolute encoder calibration positions when main 400V power is switched off. CRITICAL REQUIREMENT: The battery pack must be replaced every 24 months or immediately when Warning LED D14 illuminates yellow. WARNING: Always replace the battery pack with MAIN POWER TURNED ON. If the battery is disconnected while main power is OFF, all joint encoder zero offsets will be permanently cleared from non-volatile memory, requiring a complete Axis Zero Calibration.

3. Mechanical Calibration & Axis Zeroing

Axis zero calibration aligns the mechanical kinematic model with internal optical encoder registers. Recalibration is mandatory following servo motor replacement, timing belt retensioning, or mechanical collision.

3.1 Mechanical Locking Pin Sequence

Step 1: Switch controller to MANUAL T1 mode (velocity capped at 250 mm/s).
Step 2: Insert 8.0 mm ground calibration pin (Tool Part #CAL-PIN-08) into Axis 2 alignment borehole until fully seated.
Step 3: Jog Axis 2 in joint coordinates until the pin slides freely through both the fixed link and the rotating casting.
Step 4: Repeat the alignment sequence for Axis 3 using the 10.0 mm calibration pin (#CAL-PIN-10).
Step 5: On the Teach Pendant, navigate to Menu -> Calibration -> Zero Calibration and confirm the offset write sequence.

3.2 Joint Range Limits & Maximum Speeds

Joint Axis	Motion Range	Maximum Joint Speed	Gearbox Type
Axis 1 (Base Rotation)	-170° to +170°	180° / s	RV Cycloidal Gearbox
Axis 2 (Lower Arm)	-100° to +135°	150° / s	RV Cycloidal Gearbox
Axis 3 (Upper Arm)	-210° to +65°	180° / s	RV Cycloidal Gearbox
Axis 4 (Wrist Roll)	-200° to +200°	320° / s	Harmonic Drive CSF-25
Axis 5 (Wrist Pitch)	-125° to +125°	320° / s	Harmonic Drive CSF-25
Axis 6 (Wrist Twist)	-360° to +360°	460° / s	Harmonic Drive CSF-20

4. Diagnostic Fault Codes & Troubleshooting Matrix

When the robot controller detects an abnormal state, an alarm code appears on the Teach Pendant display. Refer to the table below for root causes and designated corrective actions.

Error Code	Fault Description	Root Cause	Corrective Action
Error E-301	Over-current on Joint 3	Mechanical binding or motor winding sho	Check Axis 3 gearbox backlash. Measure phase resistanc
Error E-405	Resolver Comm Timeout	Broken shielding on encoder cable harness	Inspect flexible cable track bundle at Joint 2. Replace cabl
Error E-512	DC Bus Under-voltage	Main supply dropped below 340V AC	Verify facility 3-phase input supply with digital multimeter.
Error E-702	Transformer Thermal Overload	Cabinet internal temp exceeded 55°C	Transformer thermal switch tripped at 115°C. Clean cabine
Error E-819	Collision Detection Trigger	Torque feedback exceeded allowable enve	Check for physical obstacle in workspace. Clear obstructio

5. Preventive Maintenance & Lubrication Schedule

Strict adherence to preventive maintenance intervals ensures kinematic accuracy and prevents unexpected downtime.

Operating Interval	Inspection Item	Lubricant / Part Spec	Standard Criterion
Every 500 Hours	Mechanical brake holding torque	N/A	Brake must hold rated payload with 150% torque ma
Every 2,500 Hours	Axis 1-3 RV gear grease	Kyodo Yushi VIGOGREASE RE0	Extract 50 ml sample, inspect for metallic particulate
Every 5,000 Hours	Axis 4-6 Harmonic drive grease	Mobilith SHC 100 Synthetic	Purge old grease through drain port until fresh greas
Every 10,000 Hours	Joint 4 timing belt tension	Belt Spec: 450-5GT-15	Measure natural frequency: 180 N tension (± 15 N).
Every 24 Months	Encoder Memory Battery Pack	BAT-36-P200 (Li-SOCl2)	Mandatory replacement while controller is energized

5.1 Special Lubricant Handling Notice

CAUTION: Never mix VIGOGREASE RE0 with standard lithium complex petroleum greases. Incompatible chemical thickeners cause grease liquefaction and catastrophic tooth spalling inside the cycloidal reduction stages.